

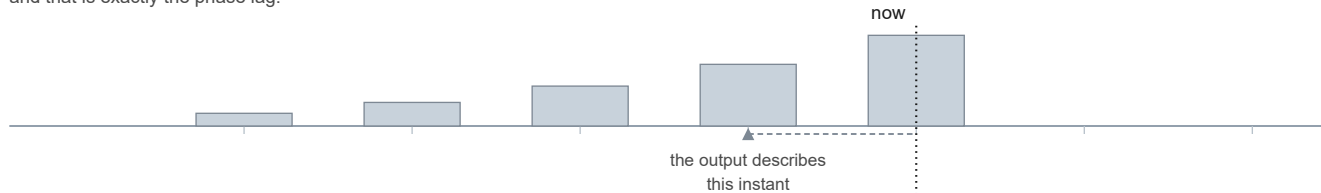
Three ways to smooth a command, and what each one costs

The kernel is the same shape throughout. Only where it sits in time changes.

Causal low-pass

weights decay into the past. The centre of mass sits behind now, and that is exactly the phase lag.

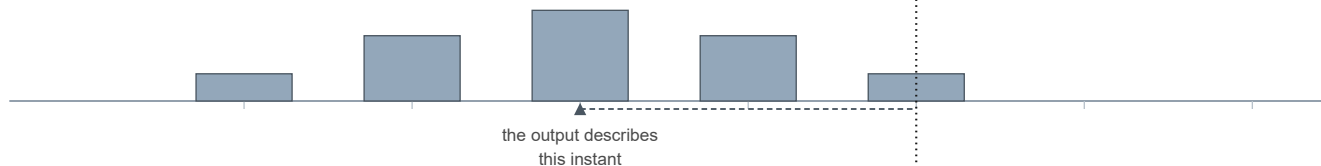
 sample that has not happened yet



The centre of mass sits about a step behind. Costs 25 points of motion (Section 5.3).

Zero phase by delaying

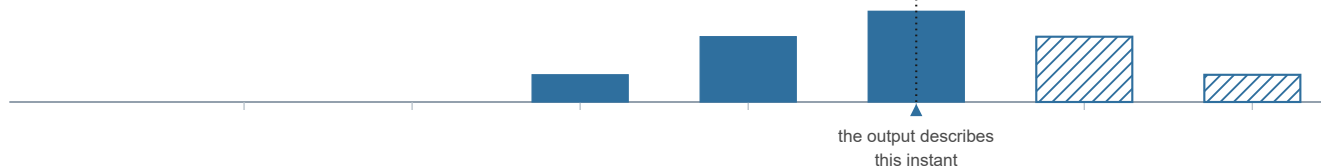
symmetric weights, so no phase distortion, but they are centred on a sample that is already two steps old.



40 ms of added latency. Falls at speed, up to every episode (Section 5.5).

Predictive, this paper

symmetric weights centred on now. The right half does not exist yet, so a small per-joint model estimates it.



No lag and no latency. Bounded by how predictable the command is.

